NR2DT-03: Step 3 — Design

> **Series:** NR2DT — New Relic to Dynatrace Migration Steps | **Notebook:** 3 of 10 | **Created:** April 2026 | **Last Updated:** 11/03/2026

Overview

Goal of this step: lock in the target Dynatrace architecture before any artifact migrates. Bucket strategy, host group taxonomy, IAM model, and OpenPipeline routing are all decided here.

Procedural runbook — see **FAQ-01** (host group naming strategy) and **ORGNZ-02 / ORGNZ-99** (Grail bucket strategy) for the recommended design patterns this step uses.

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step

Requirement	Details
Completed	NR2DT-02 — Strategize
Format	Procedural step — use as a runbook; defer to NRLC for depth
Topic deep dives	FAQ-01 (host group naming), ORGNZ-02 / ORGNZ-99 (bucket strategy), IAM-04/05 (policy design), IAM-11 (WORKSHOP)

1. What You'll Produce

Artifact	Format	Purpose
`bucket-design.md`	Markdown	Bucket names, retention, pricing model
`host-group-design.md`	Markdown	Host group taxonomy and naming
`iam-model.md`	Markdown	Groups, policies, scopes
`openpipeline-design.md`	Markdown	Enrichment + routing + filter rules
`terraform/`	HCL	Phase 1 Terraform module skeleton

2. Bucket Design

Use the recommended pattern from **ORGNZ-02 — Understanding Grail Buckets** (with the consolidated reference in **ORGNZ-99**) as your starting template:

Bucket	Retention	Pricing	Purpose
`<org>_app_logs`	30 days	Retain w/ Included Queries	Application logs
`<org>_infra_logs`	14 days	Retain w/ Included Queries	Infrastructure / K8s
`<org>_security_logs`	365 days	Usage-based	Audit trail / PCI
`<org>_spans`	14 days	Usage-based	Distributed traces
`<org>_events`	90 days	Usage-based	Platform events
`<org>_bizevents`	90 days	Usage-based	Business events

Retain with Included Queries is a Log Management & Analytics option — *"Customers can split a log bucket's retention period into two parts"* — and covers only the first 10–35 days (*"Retain with Included Queries is available only for the first 10–35 days of retention."*). The DPS capabilities page lists no included-queries option for Traces or Events — only separate *Retain* and *Query* capabilities — so keep span, event and business-event buckets usage-based.

> **Sources:** Log Management & Analytics (DT docs), Retain with Included Queries (DT docs), DPS capabilities (DT docs).

Use `<org>_` as the prefix — it lets IAM policies scope with one `storage:bucket-name startsWith "<org>_"` condition.

Document any deviations from this template (e.g., per-app buckets) and why.

3. Host Group Design

Per **FAQ-01 — Host Group Naming Strategy**, group hosts by ownership boundary, not by team alone. Common dimensions:

- Environment (`prod-app` , `nonprod-app` , `prod-infra`)
- Compliance scope (`payment-pci`)
- Workload tier (`prod-batch` , `prod-stream`)

Plan a host group per (environment × ownership) combination. Avoid a single all-hosts group — it breaks per-environment thresholds, IAM scoping, and automation safety.

4. IAM Model

Recommended starting groups:

Group	Bucket Access
`<org>-admins`	All `<org>_` buckets
`<org>-sre`	All operational: app + infra logs, spans, events
`<org>-app-eng`	App logs only

Group	Bucket Access
`<org>-readonly`	All `<org>_` buckets except security

Each statement grants a Grail table permission (`storage:logs:read` and so on) together with `storage:buckets:read`, scoped by the `storage:bucket-name` condition. The bucket permission is needed as well as the table permission — the IAM reference describes `storage:buckets:read` as *"Required additionally to a table permission."*

```
// &lt;org&gt;-admins
ALLOW storage:buckets:read, storage:logs:read, storage:spans:read,
storage:events:read, storage:bizevents:read
  WHERE storage:bucket-name startsWith "&lt;org&gt;_";

// &lt;org&gt;-sre
ALLOW storage:buckets:read, storage:logs:read WHERE storage:bucket-name IN
("&lt;org&gt;_app_logs", "&lt;org&gt;_infra_logs");
ALLOW storage:buckets:read, storage:spans:read WHERE storage:bucket-name =
"&lt;org&gt;_spans";
ALLOW storage:buckets:read, storage:events:read WHERE storage:bucket-name =
"&lt;org&gt;_events";

// &lt;org&gt;-app-eng
ALLOW storage:buckets:read, storage:logs:read WHERE storage:bucket-name =
"&lt;org&gt;_app_logs";

// &lt;org&gt;-readonly — security bucket excluded from the ALLOW scope
ALLOW storage:buckets:read, storage:logs:read, storage:spans:read,
storage:events:read, storage:bizevents:read
  WHERE storage:bucket-name startsWith "&lt;org&gt;_" AND storage:bucket-name
!= "&lt;org&gt;_security_logs";
```

Do not exclude the security bucket with a conditional DENY on a table permission. It does not do what it reads as: *"Conditional DENY statements involving Grail table permissions are not supported and are executed as unconditional DENY."* A `DENY storage:logs:read WHERE storage:bucket-name = "<org>_security_logs";` bound to the SRE, app-eng or readonly group would remove **all** log access for that group. Exclude the bucket from the ALLOW scope instead, as `<org>-readonly` does above. If you do need a DENY, the only storage permission that accepts a conditional one is the bucket permission: *"Conditional DENY statements are supported only for the buckets:read permission."*

For policy design in depth (parameterized policies, boundaries), see **IAM-04** and **ORGNZ**.

> **Sources:** IAM policy statement syntax (DT docs), IAM policy reference (DT docs), read 10/05/2026.

5. OpenPipeline Layout

Routing rules in priority order (first match wins):

Priority	Matcher	Bucket
1	`matchesValue(log.source, "*audit*") OR loglevel == "CRITICAL"`	`<org>_security_logs`
2	`in(k8s.namespace.name, {"kube-system", "kube-public", ...})`	`<org>_infra_logs`
3	App namespace match	`<org>_app_logs`
4	catch-all	`default_logs` (review weekly)

Enrichment rules for any NR workload identifier (replaces NR's tag/workload model):

```
matcher: matchesValue(k8s.namespace.name, "prod-*")   # *prod* would also match nonprod-* and preprod-*
field: environment
value: prod
```

Drop rules for cost optimization (decide which apply now vs. Wave 5):

- Drop health-check logs (`matchesValue(content, "*GET /health*")`)
- Drop DEBUG logs in production
- Reduce sidecar proxy logs at the source or forwarder (OpenPipeline has no log sampling processor)

6. Step Exit Criteria

S3 — Architecture Locked

- [ ] `bucket-design.md` final; bucket names + retention + pricing model documented

- [ ] `host-group-design.md` final; taxonomy understood by SRE + service owners
- [ ] `iam-model.md` final; security stakeholder signed off
- [ ] `openpipeline-design.md` final; routing rules sequenced
- [ ] Phase 1 Terraform module skeleton committed (will be applied in Wave 0 of Step 5)

Next step: NR2DT-04 — Translate (NRQL→DQL run; manual triage).

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